AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 15 · STUDENT LESSON

Training and validation curves

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Why does the lowest training loss in a learning-rate experiment not establish the best performance on new examples?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Compute comparable training and validation losses at saved checkpoints. Try the recall before opening notes.

Your learning target

I can compute comparable **training** and validation losses at saved **checkpoint**s.

I can choose among observed **checkpoint**s using a declared validation rule.

I can distinguish development **evidence** from final test **evidence**.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Training** a small network. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Compute comparable **training** and validation losses at saved **checkpoint**s

Choose among observed **checkpoint**s using a declared validation rule

Distinguish development **evidence** from final test **evidence**

Retrieve what you already know

> Why does the lowest **training** loss in a learning-rate experiment not establish the best performance on new examples?

My starting explanation

This week's end goal

By the end, I can explain **training** and **validation curve**s using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Training and validation curves case. Separate observations, given facts and assumptions before choosing an explanation.

Keep the restricted network h = ReLU(x), y = vh, with only v trained. **Training** pairs remain (1, 2) and (2, 4). Start v = 0 and make four full-data updates at rate 0.1. Save states v = 0, 1, 1.5, 1.75 and 1.875.

At each saved state, evaluate both splits without changing v during evaluation.

For this deliberately constructed demonstration, validation pairs are (3, 4.5) and (4, 6). Their targets follow a different slope from the **training** pairs.

This makes the two trends disagree visibly; it is not **evidence** that these few rows represent a real population. Validation labels do not enter the gradient.

Updates	v	Training MSE	Validation MSE
0	0	10	28.125
1	1	2.5	3.125
2	1.5	0.625	0
3	1.75	0.15625	0.78125
4	1.875	0.0390625	1.7578125

Both columns use ordinary MSE in the same target units, evaluated on their own fixed rows. **Training** MSE is 2.5(v - 2) squared; validation MSE is 12.5(v - 1.5) squared.

Different **input** values and target rules affect the loss levels. A gap alone does not identify its cause.

List the given facts and one fact that is not supplied.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Training and validation curves. Connect each labelled input, deciding step and result.

For **checkpoint** 3, v = 1.75. **Training prediction**s are [1.75, 3.5], residuals [-0.25, -0.5], and MSE 0.15625. Validation **prediction**s are [5.25, 7], residuals [0.75, 1], and MSE 0.78125.

All four **prediction**s must use the same saved value.

```
def mse(v, rows):
    if not rows:
        raise ValueError("Need evaluation rows")
    return sum((v * max(0.0, x) - target) ** 2
               for x, target in rows) / len(rows)

train = [(1.0, 2.0), (2.0, 4.0)]
valid = [(3.0, 4.5), (4.0, 6.0)]
checkpoints = [0.0, 1.0, 1.5, 1.75, 1.875]
validation_losses = []
for step, v in enumerate(checkpoints):
    validation_losses.append(mse(v, valid))
    print(step, mse(v, train), mse(v, valid))
best = min(range(len(checkpoints)),
           key=lambda i: (validation_losses[i], i))
print("Selected checkpoint", best)  # 2
```

The code evaluates supplied **checkpoint**s; it does not train on validation data. It assumes finite numeric pairs and selects the lowest observed validation MSE, breaking exact ties in favour of the earlier **checkpoint**.

The rule is chosen before inspecting the values.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Training and validation curves. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Observed validation MSE at **checkpoint**s 0 to 4 is [5, 2, 1, 1, 1.2]. Lowest loss wins with earlier exact ties. Which **checkpoint** is selected? What if **training** loss is lowest at **checkpoint** 4?

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Epoch 3 validation loss is instead 1.5.

Predicted result

Prediction: Select epoch 3 under the unchanged criterion.

Why the result follows

Reason: Its measured validation loss is now below epoch 2′s2; the **training curve** alone never settled selection.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Training and validation curves to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Checkpoint	Saved model parameters and the information needed to identify a training state
Training curve	A record of a training measure at successive saved states
Validation curve	A record of a development measure on separate examples at those states
Selection rule	A declared procedure for choosing among candidate models or checkpoints

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Training and validation curves, predict a step, then compare with its explanation.

Later updates improve only one split

Series	2	3	4
Training solid line	0.625	0.15625	0.0390625
Validation dashed line	0	0.78125	1.7578125

This chart shows checkpoints 2 to 4 only, on one MSE scale from 0 to 2. Circles and a solid line mark training; squares and a dashed line mark validation. Lines connect measured checkpoints, not additional evaluations. The full table includes checkpoints 0 and 1.

Using the table, calculate the training and validation loss changes from checkpoint 2 to checkpoint 4. Which saved parameter does the selection rule choose?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Training and validation curves

Plot **training** and validation loss for **checkpoint**s before opening final cases.

The facts we will use

Supplied facts

Supplied **checkpoint** losses: epoch 1 train 4, val 3; epoch 2 train 2, val 2; epoch 3 train 1, val 2.5. Selection: smallest validation loss.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/15

Checkpoint losses

	training	validation
epoch 1	4	3
epoch 2	2	2
epoch 3	1	2.5

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/15

Training and validation curves

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read training curve

Training loss falls 4 -> 2 -> 1.



2. Read validation curve

Validation loss falls 3 -> 2 then rises to2.5.



3. Choose checkpoint

Select epoch 2, not epoch 3, under the declared validation criterion.

CHANGE ONE THING

Epoch 3 validation loss is instead 1.5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read training curve	Training loss falls 4→2→1.
2. Read validation curve	Validation loss falls 3→2 then rises to2.5.
3. Choose checkpoint	Select epoch 2, not epoch 3, under the declared validation criterion.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The **training curve** measures fit to examples used for learning. The **validation curve** measures a separate set used to choose a **checkpoint**. The final **training** loss is smallest here, but its validation loss has risen.

Under the supplied **selection rule**, keep **checkpoint** two. A later final test must remain outside both learning and **checkpoint** selection.

What this example does not establish

Keep the stated limit

A single curve has sampling and run variability; validation-selected **checkpoint**s still require untouched final **evidence**.

Keep the stated limit

Read the labelled picture: Can the original epoch 3 **training** loss 1 justify claiming best generalisation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Epoch 3 validation loss is instead 1.5. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Training and validation curves claim and explain a repair using the supplied evidence.

Check the claim against the facts

Falling **training** loss with rising validation loss can be consistent with over**fitting**, but it is not a complete diagnosis. In this example the split-specific target rules are known to differ.

In another dataset, inspect representativeness, pre**process**ing, leakage, label quality and evaluation settings before attributing the pattern to model capacity alone.

Check the claim against the facts

Evaluate comparable quantities: do not compare a **training** sum with a validation mean, or a **training** objective containing a penalty with unpenalised validation MSE. Keep the row sets fixed across **checkpoint**s.

> Do not mix losses recorded before each batch update with a full-split evaluation of the final **checkpoint**.

Locate the earliest unsupported step

A **training** report shows sum of squared errors 12 over six rows. Validation shows mean squared error 3 over two rows in the same target units. A learner says **training** error is four times larger. Repair the comparison.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Training and validation curves with support, then explain the deciding step in your own words.

At v = 1, compute both validation **prediction**s, residuals and MSE on (3, 4.5) and (4, 6).

Using the table, calculate the training and validation loss changes from checkpoint 2 to checkpoint 4. Which saved parameter does the selection rule choose?

Explain your prediction to a partner before comparing results.

Practise with less help: Use the declared rule

Use workbook W2 for the connected example "**Training** and **validation curve**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Training** and **validation curve**s. Proposed explanation: "The last **checkpoint** is always best". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Can the original epoch 3 **training** loss 1 justify claiming best generalisation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Training and validation curves. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Observed validation MSE at **checkpoint**s 0 to 4 is [5, 2, 1, 1, 1.2]. Lowest loss wins with earlier exact ties. Which **checkpoint** is selected? What if **training** loss is lowest at **checkpoint** 4?

```
def mse(v, rows):
    if not rows:
        raise ValueError("Need evaluation rows")
    return sum((v * max(0.0, x) - target) ** 2
               for x, target in rows) / len(rows)

train = [(1.0, 2.0), (2.0, 4.0)]
valid = [(3.0, 4.5), (4.0, 6.0)]
checkpoints = [0.0, 1.0, 1.5, 1.75, 1.875]
validation_losses = []
for step, v in enumerate(checkpoints):
    validation_losses.append(mse(v, valid))
    print(step, mse(v, train), mse(v, valid))
best = min(range(len(checkpoints)),
           key=lambda i: (validation_losses[i], i))
print("Selected checkpoint", best)  # 2
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Training and validation curves. Explain why each step is justified.

Explain → apply → challenge

At v = 1, compute both validation **prediction**s, residuals and MSE on (3, 4.5) and (4, 6).

A **training** report shows sum of squared errors 12 over six rows. Validation shows mean squared error 3 over two rows in the same target units. A learner says **training** error is four times larger. Repair the comparison.

Training loss falls while validation loss rises. Give two checks before declaring model complexity the cause, and explain why repeatedly selecting with validation makes it development **evidence**.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled **training** and **validation curve**s **evidence** sheet to Image-pattern Network Investigation. Show how you can compute comparable **training** and validation losses at saved **checkpoint**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Restore **checkpoint** 2, v = 1.5, if following the declared rule. Its zero validation MSE is a result on two constructed development examples. Choosing this **checkpoint** using those examples makes them part of model selection.

After all choices are frozen, a separate untouched test set can provide final **evidence** within its own coverage limits.

> **A team calls checkpoint 2 independently proven because its validation MSE is zero, then uses the test results to choose another checkpoint. Identify both reporting problems and propose a clean procedure.**

Next week adds a weight penalty and an early-stopping policy, keeping their roles distinct.

Training loss falls while validation loss rises. Give two checks before declaring model complexity the cause, and explain why repeatedly selecting with validation makes it development **evidence**.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Training and validation curves case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Later updates improve only one split

Series	2	3	4
Training solid line	0.625	0.15625	0.0390625
Validation dashed line	0	0.78125	1.7578125

This chart shows checkpoints 2 to 4 only, on one MSE scale from 0 to 2. Circles and a solid line mark training; squares and a dashed line mark validation. Lines connect measured checkpoints, not additional evaluations. The full table includes checkpoints 0 and 1.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Evaluate new checkpoints independently

Use h = ReLU(x), y = vh, candidate v values 1, 1.5 and 2. **Training** pairs are (1, 2), (2, 4); validation pairs are (1, 1), (2, 2). Calculate both MSE columns and select by minimum validation MSE.

Explain the disagreement with selection by **training** loss.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Use h = ReLU(x), y = vh, candidate v values 1, 1.5 and 2. **Training** pairs are (1, 2), (2, 4); validation pairs are (1, 1), (2, 2). Calculate both MSE columns and select by minimum validation MSE.

Explain the disagreement with selection by **training** loss.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Training and validation curves, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Plot **training** and validation loss for **checkpoint**s before opening final cases.

By the end, I can	How I will show it
Compute comparable training and validation losses at saved checkpoints	Explain the deciding step, show a trace or test, and state one limit.
Choose among observed checkpoints using a declared validation rule	Explain the deciding step, show a trace or test, and state one limit.
Distinguish development evidence from final test evidence	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Training and validation curves without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Compute comparable **training** and validation losses at saved **checkpoint**s

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Plot **training** and validation loss for **checkpoint**s before opening final cases.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Select epoch 2, not epoch 3, under the declared validation criterion.

1. Read **training curve**

2. Choose **checkpoint**

3. Read **validation curve**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Choose **checkpoint** Select epoch 2, not epoch 3, under the declared validation criterion. The **training curve** measures fit to examples used for learning.

The **validation curve** measures a separate set used to choose a **checkpoint**. The final **training** loss is smallest here, but its validation loss has risen. Under the supplied **selection rule**, keep **checkpoint** two.

A later final test must remain outside both learning and **checkpoint** selection.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Epoch 3 validation loss is instead 1.5. Which conclusion is supported by the supplied facts?

1. Select epoch 3 under the unchanged criterion.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Select epoch 3 under the unchanged criterion. Its measured validation loss is now below epoch 2′s2; the **training curve** alone never settled selection.

Repair a missing building block

Review Learning rate experiments (ai-5/14). Goal: Run a controlled comparison of learning rates. Short practice: What should a fair rate comparison hold fixed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Training** a small network (ai-5/12). Goal: Trace repeated forward loss gradient and update steps. Short practice: Which parameter changed, and which part was not trained?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Slopes and local change (ai-5/9). Goal: Calculate an average rate of change over a stated interval. Short practice: Which calculation disproves “any increase helps”?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

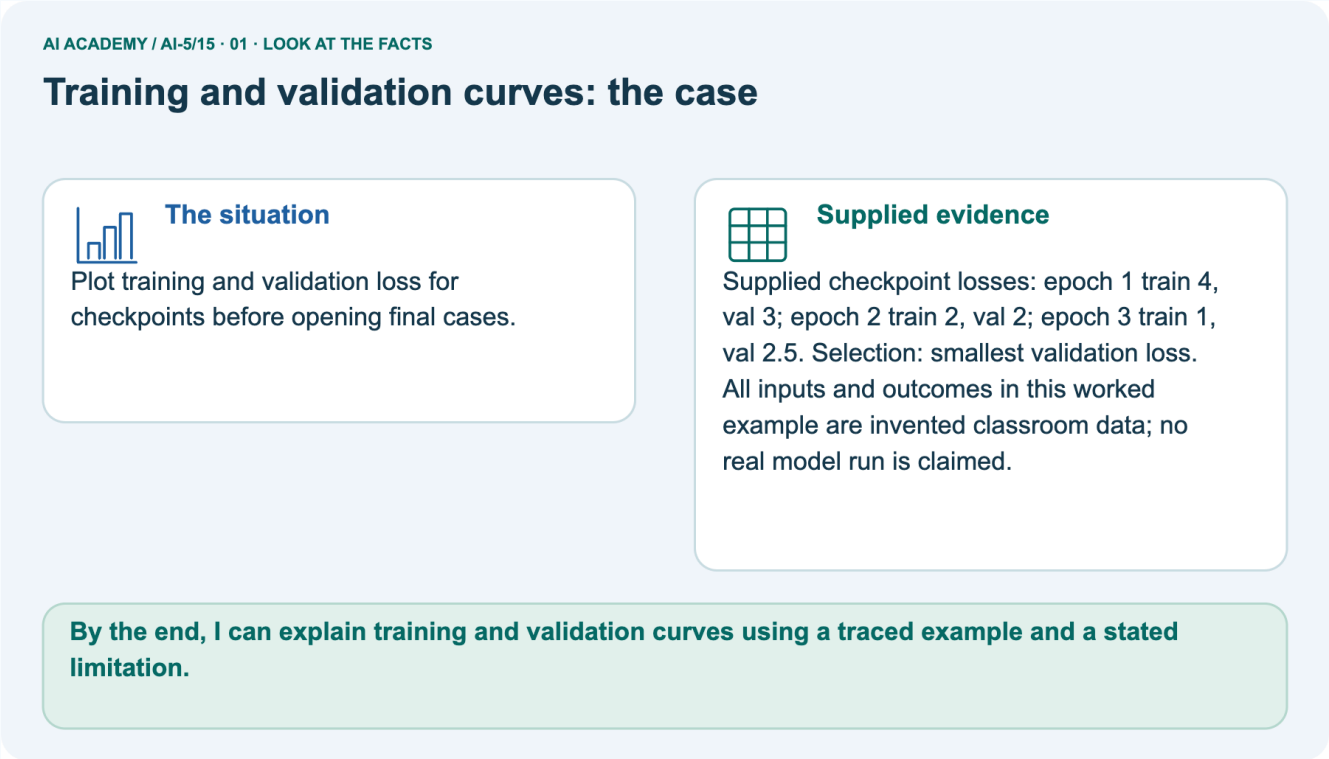
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Training and validation curves graphic using its labels, deciding evidence and relevant limitation.

01. Training and validation curves: the case



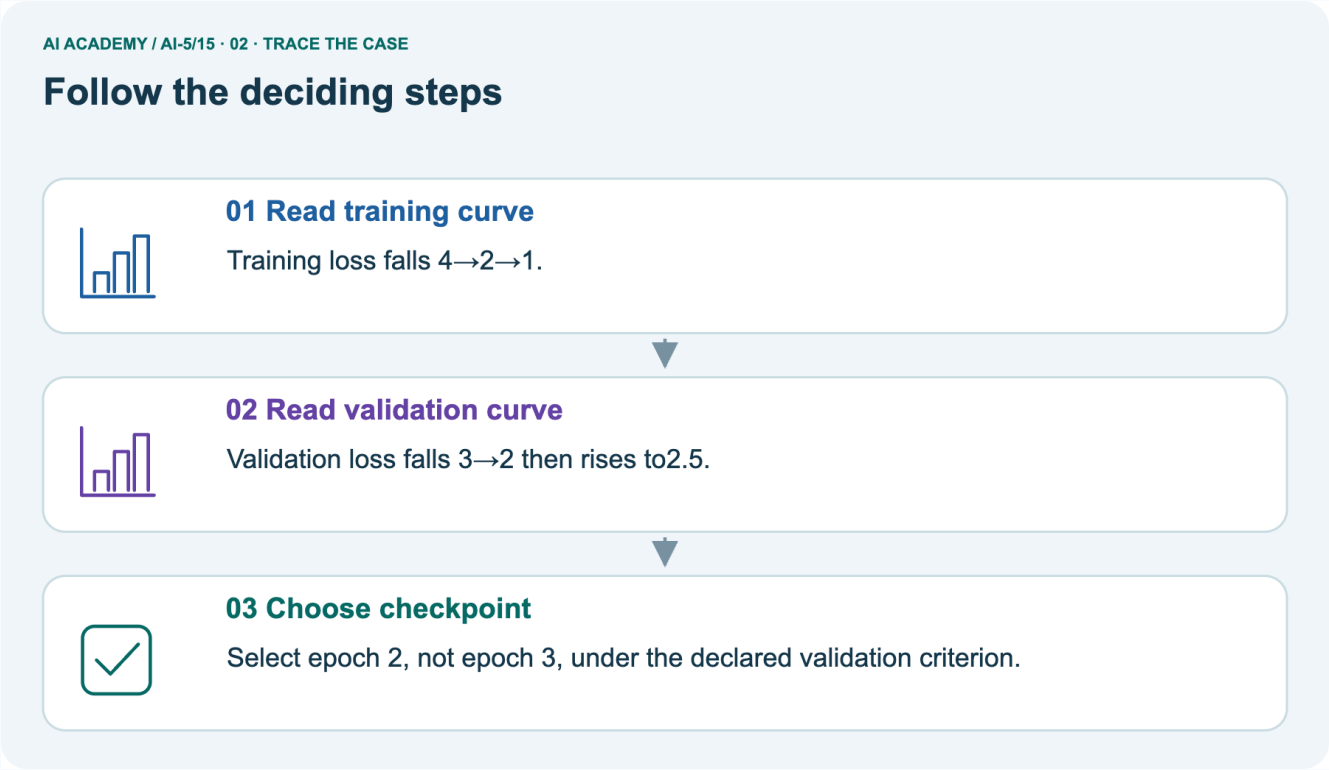
Follow the labelled picture with the matching explanation. Open the vector diagram

Plot **training** and validation loss for **checkpoint**s before opening final cases. Supplied facts: Supplied **checkpoint** losses: epoch 1 train 4, val 3; epoch 2 train 2, val 2; epoch 3 train 1, val 2.5. Selection: smallest validation loss.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read **training curve**: **Training** loss falls 4→2→1. → Read **validation curve**: Validation loss falls 3→2 then rises to2.5. → Choose **checkpoint**: Select epoch 2, not epoch 3, under the declared validation criterion.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Checkpoint losses

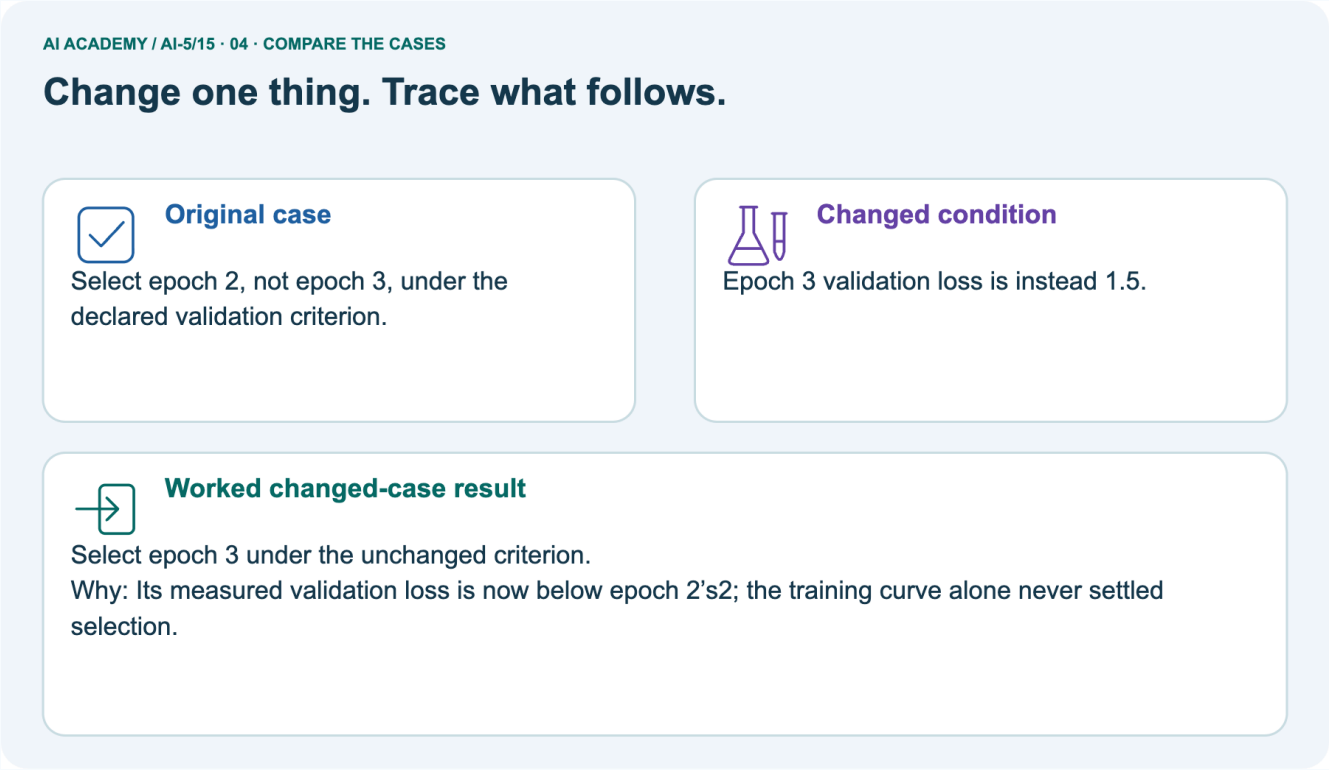


Follow the labelled picture with the matching explanation. Open the vector diagram

Checkpoint losses. epoch 1: **training** = 4, validation = 3. epoch 2: **training** = 2, validation = 2. epoch 3: **training** = 1, validation = 2.5.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



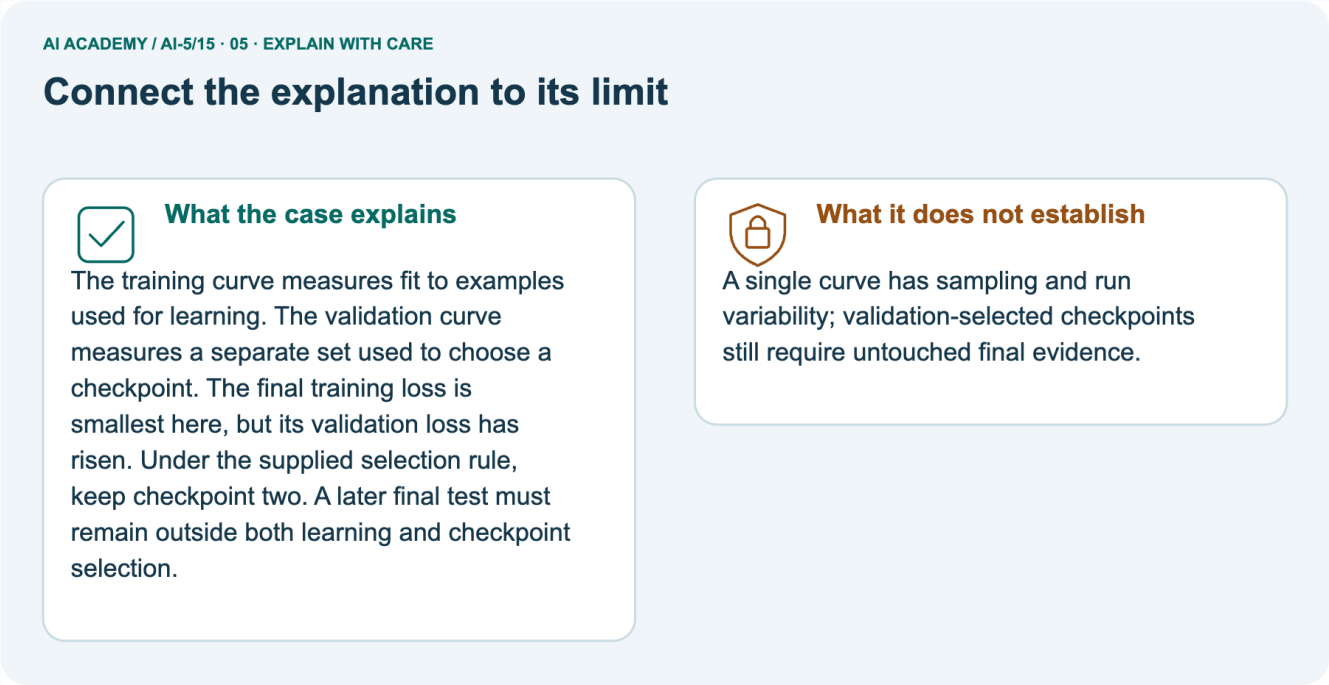
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Select epoch 2, not epoch 3, under the declared validation criterion. Changed condition: Epoch 3 validation loss is instead 1.5. Expected result: Select epoch 3 under the unchanged criterion.

Reason: Its measured validation loss is now below epoch 2's2; the **training curve** alone never settled selection.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

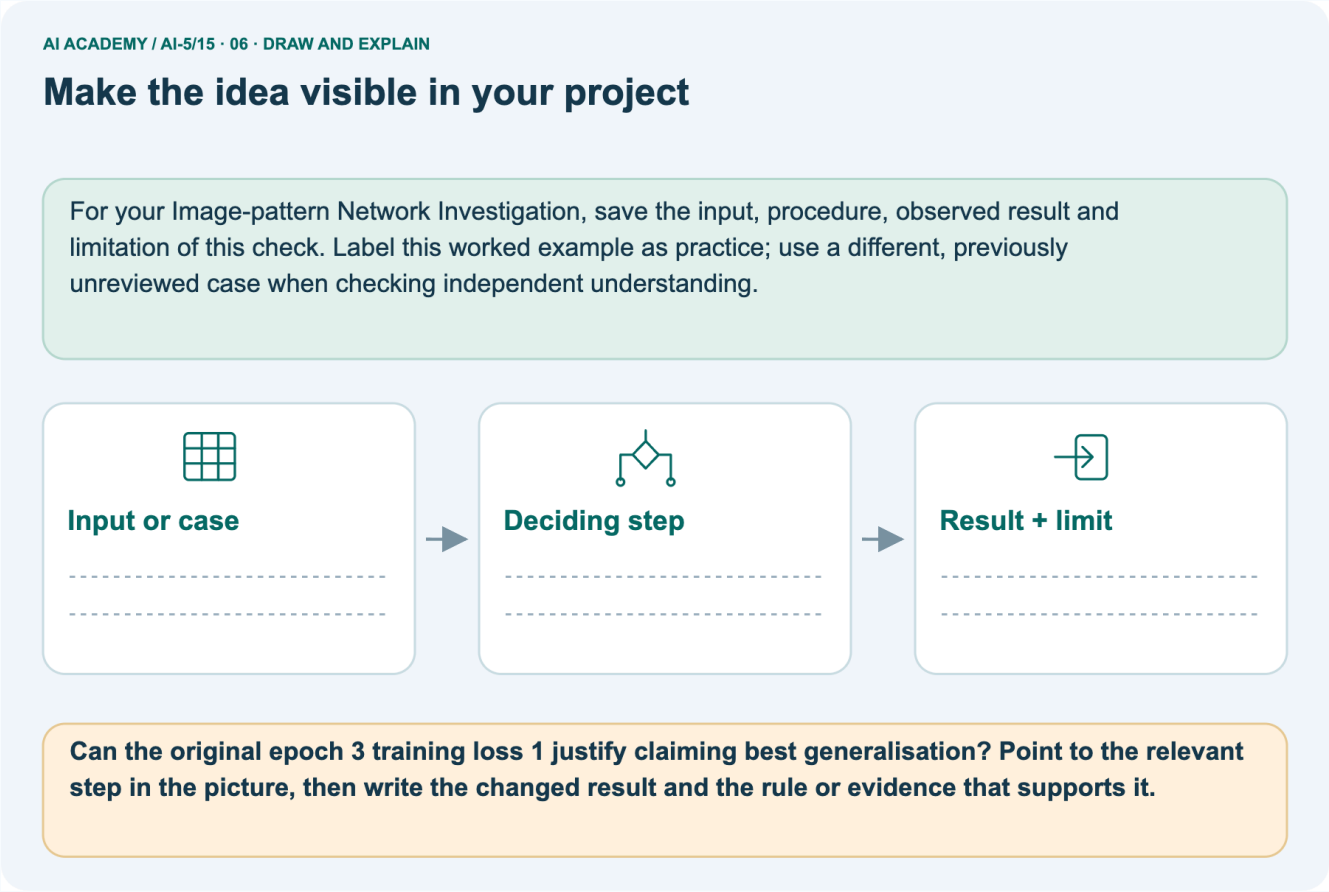
The **training curve** measures fit to examples used for learning. The **validation curve** measures a separate set used to choose a **checkpoint**. The final **training** loss is smallest here, but its validation loss has risen.

Under the supplied **selection rule**, keep **checkpoint** two. A later final test must remain outside both learning and **checkpoint** selection.

Limit: A single curve has sampling and run variability; validation-selected **checkpoint**s still require untouched final **evidence**.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can the original epoch 3 **training** loss 1 justify claiming best generalisation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.